



Rincoin Whitepaper

S1 Monetary Schedule Candidate

Rincoin Community Forge

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 Candidate Discussion Draft

Document version 1.0

This candidate shows how the Rincoin whitepaper would define the network under the S1 monetary schedule. It is not adopted consensus and remains subject to public review, compatible implementation, testing, and voluntary network adoption.

Abstract

Rincoin is a decentralised Layer-1 UTXO cryptocurrency secured by proof of work. It combines direct ownership and independent validation with a 60-second target block interval, the memory-hard RinHash proof-of-work construction, and Dark Gravity Wave v3 difficulty adjustment. Its monetary rule is deterministic: the maximum permitted block subsidy depends only on block height.

Rincoin began with a 50 RIN subsidy and 210,000-block reward epochs. The first four epochs used binary halving. From block height 840,000, the maximum subsidy follows a recursive 19/20 rule, nominally a 5% reduction, with integer-floor rounding after each epoch. There is no subsidy floor or perpetual tail; rounding eventually reduces subsidy to zero. The complete schedule permits at most 44624993.01750000 RIN.

The network has no premine or developer tax as stated in the original RinCoin whitepaper ([RinCoin Whitepaper 2025](#)). Its operation depends on independently validating nodes, miners, wallets, and services choosing compatible software. RinHash seeks to increase the cost of fixed-function mining specialisation, but does not guarantee equal hardware access. Long-term security depends on subsidy, fees, coin value, liquidity, mining efficiency, and broad operational participation.

1 Rincoin

Rincoin is a proof-of-work digital currency for direct settlement and self-custody. It uses the UTXO ledger model and cumulative-proof-of-work chain selection. Independent full nodes validate transactions and blocks; miners order transactions by extending the valid chain with the greatest cumulative work ([Rin-coin contributors 2026](#)).

The design combines one-minute target blocks with a memory-hard proof-of-work function and finite, predictable issuance. It favours a compact base layer, transparent validation rules, open-source software, and direct control of private keys.

Parameter	Value
Ledger model	Layer-1 UTXO
Consensus mechanism	Proof of work
Proof-of-work function	RinHash
RinHash construction	BLAKE3 -> Argon2d -> SHA3-256
Difficulty adjustment	Dark Gravity Wave v3
Target block interval	60 seconds
Initial block subsidy	50 RIN
Historical reward epoch	210,000 blocks
Blocks 0-839,999	Binary integer halving, nominally 50%, with base-unit floor rounding
Blocks 840,000 and later	Recursive 19/20 rule every 210,000 blocks; nominally 5%, with per-epoch integer-floor rounding
Subsidy floor	None
Maximum scheduled issuance	44624993.01750000 RIN
Historical upper bound stated in the original whitepaper	168,000,000 RIN
Premine	None stated in the original whitepaper
Developer tax	None stated in the original whitepaper

2 Ledger and Transactions

Rincoin transactions consume existing unspent transaction outputs and create new outputs. Spending authority is evaluated under the applicable signature and script rules. During block validation, full nodes verify that referenced inputs are available and value-conserving and perform input-script checks before updating the UTXO set ([Rin-coin contributors 2026](#)).

Each block contains a coinbase transaction that may claim no more than the height-dependent subsidy plus transaction fees. Fees transfer existing RIN to miners and are not new issuance. If a miner claims less than the maximum permitted subsidy, the difference is not reissued ([Rin-coin contributors 2026](#)).

3 Blocks, Proof of Work and Difficulty

Miners assemble valid transactions into candidate blocks and search for a header whose proof-of-work hash satisfies the current target. RinHash combines BLAKE3, Argon2d, and SHA3-256. Its memory-hard stage is intended to increase the cost of fixed-function specialisation compared with simple hash-only proof-of-work designs. Specialisation resistance is parameter-dependent and is not a permanent guarantee of equal CPU, GPU, FPGA, or ASIC economics ([Rin-coin contributors 2026](#)).

Rincoin targets one block every 60 seconds. Dark Gravity Wave v3 adjusts difficulty from recent block history so that the network can respond to changes in available hashrate. Difficulty adjustment governs the work required for a valid block; it does not alter the subsidy schedule ([Rin-coin contributors 2026](#)).

Nodes choose among valid histories by cumulative proof of work. A chain with more blocks is not necessarily preferred if it represents less accumulated work.

4 Monetary Policy

The maximum permitted subsidy is a deterministic function of block height. The schedule uses integer base units, where one RIN equals 100,000,000 base units. It does not depend on market or ledger state ([Rin-coin contributors 2026](#)).

Rincoin began with a 50 RIN block subsidy. For blocks below height 840,000, the subsidy follows the deployed binary-halving rule across 210,000-block epochs: 50, 25, 12.5, and 6.25 RIN ([Rin-coin contributors 2026](#)).

From block height 840,000, the maximum permitted subsidy follows a recursive 19/20 rule, nominally a 5% reduction, with integer-floor rounding after each 210,000-block epoch. The first post-840k subsidy is 5.93750000 RIN. Each later epoch is calculated from the preceding integer-rounded epoch subsidy by multiplying by 19, dividing by 20, and rounding down to a whole base unit. Effective reductions can differ slightly from 5% after rounding. There is no fixed subsidy phase and no perpetual tail. Repeated integer rounding eventually produces zero subsidy.

The complete height-based schedule permits at most 44624993.01750000 RIN. Final circulating or spendable supply can be lower because unclaimed subsidy is not reissued.

The original RinCoin whitepaper stated a 168,000,000 RIN maximum while also specifying a 50 RIN initial subsidy and 210,000-block binary halving. Those parameters produce a documentation-versus-consensus inconsistency: unchanged binary halving approaches approximately 21 million RIN. The 168 million value remains a historical documented upper bound, not the effective maximum of this schedule ([RinCoin Whitepaper 2025](#); [Rin-coin contributors 2026](#)).

The change at height 840,000 alters future coinbase validation and is therefore a hard-fork consensus change. It preserves all valid earlier blocks, existing UTXOs, balances, and ownership claims.

5 Mining Incentives and Network Security

Miner revenue consists of block subsidy and transaction fees. Its economic value depends on the RIN price, usable liquidity, fee demand, hardware efficiency, energy cost, and competition among miners.

The post-840k schedule replaces each future 50% subsidy cliff with a recursive 19/20 step every 210,000 blocks, nominally a 5% reduction with integer-floor rounding. This creates frequent but smaller changes in nominal subsidy. It does not guarantee stable hashrate or network security and schedules more RIN than unchanged binary halving.

As subsidy declines, security depends increasingly on transaction fees, coin value, efficient mining, and the willingness of independent operators to support the network. A nominal reward alone cannot ensure decentralisation, liquidity, or resistance to reorganisation.

6 Network Operation and Protocol Evolution

Rincoin operates through independently validating nodes, miners, wallets, exchanges, explorers, and other services using compatible rules. Validating nodes independently enforce the consensus rules implemented by the software they choose to run.

Consensus changes require publicly inspectable specifications and software, testable boundary behaviour, compatible deployments, and voluntary adoption by network participants. Publication of a draft describes a rule; it does not activate that rule.

The candidate S1 consensus arithmetic and boundary behaviour are defined in the *Rincoin 840k S1 Consensus Change Specification* ([Rincoin Community Forge 2026a](#)).

References

The candidate specification and companion review package provide the normative subsidy arithmetic, deployed-source audit, and reproducibility materials ([Rincoin Community Forge 2026b](#)).

Rin-coin contributors. 2026. *Rincoin Core: Deployed Consensus Source Snapshot*. Version b52c87778f800dc5f4e2f59c372badbc139f933f. Released March 7. <https://github.com/Rin-coin/rincoin/tree/b52c87778f800dc5f4e2f59c372badbc139f933f>.

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